

# Foreign Keys in Relational Schemas

txt2tex example

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[*RecipeId*, *IngredientId*, *EmployeeId*, *CuisineTag*, *IngredientName*, *RecipeName*, *EmployeeName*, *PositionName*]

## Pattern 1: Single-Attribute Primary Key

Each relation has exactly one pk line. txt2tex underlines the attribute in the schema box and emits a PK statement immediately after.

|                                    |
|------------------------------------|
| <u><i>Recipe</i></u>               |
| <i>recipeId</i> : <i>RecipeId</i>  |
| <i>name</i> : <i>RecipeName</i>    |
| <i>cuisine</i> : <i>CuisineTag</i> |

PK(Recipe) = {*recipeId*}

|                                           |
|-------------------------------------------|
| <u><i>Ingredient</i></u>                  |
| <i>ingredientId</i> : <i>IngredientId</i> |
| <i>name</i> : <i>IngredientName</i>       |

PK(Ingredient) = {*ingredientId*}

## Pattern 2: Composite Primary Key

Two pk lines in one schema body. Both attributes are underlined; txt2tex collapses them into one PK set: PK(RecipeIngredient) = {*recipeId*, *ingredientId*}.

|                                                  |
|--------------------------------------------------|
| <u><i>RecipeIngredient</i></u>                   |
| <u><i>recipeId</i></u> : <i>RecipeId</i>         |
| <u><i>ingredientId</i></u> : <i>IngredientId</i> |

PK(RecipeIngredient) = {*recipeId*, *ingredientId*}

## Pattern 3: Single-Attribute Foreign Key

A foreign key constraint says every value of the referencing attribute must appear in the referenced relation. No special keyword is needed: write the constraint as a Z predicate using forall/exists or projection with subset.

RecipeIngredient.recipeId must reference an existing Recipe:

$\forall ri : \text{RecipeIngredient} \bullet \exists r : \text{Recipe} \bullet ri.\text{recipeId} = r.\text{recipeId}$

Equivalently, using projection and subset:

$\text{Project}_{\text{recipeId}}(\text{RecipeIngredient}) \subseteq \text{Project}_{\text{recipeId}}(\text{Recipe})$   
 RecipeIngredient.ingredientId must reference an existing Ingredient:

$$\forall ri : \text{RecipeIngredient} \bullet \exists i : \text{Ingredient} \bullet ri.\text{ingredientId} = i.\text{ingredientId}$$

## Pattern 4: Composite FK and Self-Referential FK

Both FK constraints for RecipeIngredient can be stated together in a single zed block. A blank line between predicates produces visual separation in the output.

$$\forall ri : \text{RecipeIngredient} \bullet \exists r : \text{Recipe} \bullet ri.\text{recipeId} = r.\text{recipeId}$$

$$\forall ri : \text{RecipeIngredient} \bullet \exists i : \text{Ingredient} \bullet ri.\text{ingredientId} = i.\text{ingredientId}$$

Employee carries a managerId that references another row in the same relation. The self-referential FK constraint:

|                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Employee</i><br><i>employeeId</i> : <i>EmployeeId</i><br><i>name</i> : <i>EmployeeName</i><br><i>position</i> : <i>PositionName</i><br><i>managerId</i> : <i>EmployeeId</i> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

$\text{PK}(\text{Employee}) = \{\text{employeeId}\}$

Every managerId in Employee must match a valid employeeId in the same relation:

$$\forall e : \text{Employee} \bullet \exists m : \text{Employee} \bullet e.\text{managerId} = m.\text{employeeId}$$